

Quiz 1- Set A, B.Tech 4th Sem, GCU
Subject- Transportation Engineering- BCE23401T

Date: 18/02/2026

Roll No. _____

1. The first 20-year road development plan in India was:
 - A) Bombay Plan (1943–63)
 - B) Nagpur Plan (1943–63)
 - C) Lucknow Plan (1981–2001)
 - D) Jayakar Plan (1927)
2. The minimum width of a parking lane parallel to kerb for cars is generally:
 - A) 1.8 m
 - B) 2.2 m
 - C) 2.5 m
 - D) 3.0 m
3. For a two-way cycle track (separate from carriageway), recommended width is about:
 - A) 1.5 m
 - B) 2.0 m
 - C) 2.5 m
 - D) 3.0 m
4. During reconnaissance survey, the following is NOT studied in detail:
 - A) Obligatory points
 - B) Soil characteristics
 - C) Alignment alternatives
 - D) Traffic data (classified count)
5. Camber for cement concrete pavement is generally:
 - A) 1 in 20
 - B) 1 in 30
 - C) 1 in 50
 - D) 1 in 60
6. A vehicle moving at 80 km/h with perception-reaction time 2.5 s. Assume coefficient of friction = 0.35. Stopping sight distance on level road $\approx$
 - A) 120 m
 - B) 135 m
 - C) 150 m
 - D) 170 m
7. Maximum superelevation as per IRC in plain terrain:
 - A) 1 in 10
 - B) 1 in 12
 - C) 1 in 15
 - D) 1 in 20
8. Overtaking zone length $\geq$
 - A) 2×OSD
 - B) 3×OSD
 - C) 4×OSD
 - D) 5×OSD
9. Overtaking sight distance is based on:
 - A) 2 vehicles
 - B) 3 vehicles
 - C) 4 vehicles
 - D) 5 vehicles
10. The camber required for WBM road is approximately:
 - A) 1 in 60
 - B) 1 in 48
 - C) 1 in 33
 - D) 1 in 25

Quiz 1- Set B, B.Tech 4th Sem, GCU
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R

Roll No. _____

1. For a two-way cycle track (separate from carriageway), recommended width is about:
 - A) 1.5 m
 - B) 2.0 m
 - C) 2.5 m
 - D) 3.0 m
2. The Lucknow Plan covered period:
 - A) 1961–81
 - B) 1981–2001
 - C) 2001–2021
 - D) 1943–63
3. During reconnaissance survey, the following is NOT studied in detail:
 - A) Obligatory points
 - B) Soil characteristics
 - C) Alignment alternatives
 - D) Traffic data (classified count)
4. If $R = 250$ m, $V = 80$ km/h, required superelevation (ignore friction) $\approx$
 - A) 0.05
 - B) 0.07
 - C) 0.09
 - D) 0.11
5. For a design speed of 100 km/h, reaction time 2.5 s, SSD $\approx$ ($f=0.35$):
 - A) 160 m
 - B) 180 m
 - C) 200 m
 - D) 220 m
6. Crown height increases with:
 - A) Pavement width
 - B) Camber
 - C) Traffic
 - D) Gradient
7. Camber is provided primarily for:
 - A) Vehicle comfort
 - B) Drainage
 - C) Stability
 - D) Aesthetics
8. Maximum superelevation in snow-bound areas:
 - A) 1 in 15
 - B) 1 in 12
 - C) 1 in 10
 - D) 1 in 8
9. Road margin includes all EXCEPT:
 - A) Shoulder
 - B) Parking lane
 - C) Median
 - D) Carriageway
10. The camber for bituminous pavement ranges:
 - A) 1.5–2%
 - B) 2–3%
 - C) 3–4%
 - D) 4–5%

Quiz 1- Set C, B.Tech 4th Sem, GCU
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Roll No. _____

1. CRF (Central Road Fund) was established in:
A) 1929
B) 1934
C) 1943
D) 1956
2. For a two-way cycle track (separate from carriageway), recommended width is about:
A) 1.5 m
B) 2.0 m
C) 2.5 m
D) 3.0 m
3. Obligatory point example:
A) Agricultural land
B) River crossing
C) Forest area
D) Open plain
4. If $R = 250$ m, $V = 80$ km/h, required superelevation (ignore friction) $\approx$
A) 0.05
B) 0.07
C) 0.09
D) 0.11
5. Stopping sight distance increases when:
A) Speed decreases
B) Friction increases
C) Gradient is downward
D) Reaction time reduces
6. Formation width of 2-lane NH $\approx$
A) 7 m
B) 10 m
C) 12 m
D) 15 m
7. Camber is usually provided in shape:
A) Convex
B) Concave
C) Straight
D) Zigzag
8. For safe overtaking, minimum OSD is generally:
A) $2 \times \text{SSD}$
B) $3 \times \text{SSD}$
C) $4 \times \text{SSD}$
D) $5 \times \text{SSD}$
9. The cross slope of shoulder is usually kept:
A) Equal to camber
B) Flatter than camber
C) Steeper than camber
D) Zero
10. A raised kerb of height 15–20 cm is generally called:
A) Mountable kerb
B) Semi-barrier kerb
C) Barrier kerb
D) Submerged kerb

Quiz 1- Set D, B.Tech 4th Sem, GCU
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Roll No. _____

1. The third 20-year road plan is known as:
A) Nagpur Plan
B) Bombay Plan
C) Lucknow Plan
D) Vision 2021
2. ODR mainly connects:
A) States
B) District HQ
C) Rural areas
D) National capital
3. For a two-way cycle track (separate from carriageway), recommended width is about:
A) 1.5 m
B) 2.0 m
C) 2.5 m
D) 3.0 m
4. If $V = 60$ km/h, $f = 0.4$, $\text{SSD} \approx$
A) 70 m
B) 80 m
C) 90 m
D) 100 m
5. Super elevation is provided to counteract:
A) Skidding
B) Rolling
C) Centrifugal force
D) Friction
6. Maximum camber for earthen roads:
A) 2%
B) 3%
C) 4%
D) 5%
7. Crown height increases with:
A) Pavement width
B) Camber
C) Traffic
D) Gradient
8. The main purpose of providing kerbs is to:
A) Increase pavement thickness
B) Separate carriageway from shoulder/footpath
C) Reduce superelevation
D) Control camber
9. For a single-lane road with carriageway width 3.75 m and shoulders 1.5 m each side, roadway width equals:
A) 3.75 m
B) 5.25 m
C) 6.75 m
D) 7.50 m
10. Road margin includes all EXCEPT:
A) Shoulder
B) Parking lane
C) Median
D) Carriageway

Quiz 1- Set E, B.Tech 4th Sem, GCU
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Roll No. _____

1. IRC was formed in:
A) 1927
B) 1934
C) 1943
D) 1956
2. NH are declared under:
A) State Act
B) IRC Code
C) Parliament Act
D) CRF Act
3. Length of overtaking zone $\geq$
A) $2 \times \text{OSD}$
B) $3 \times \text{OSD}$
C) $4 \times \text{OSD}$
D) $5 \times \text{OSD}$
4. If $V=90$ km/h, $R=300$ m, e required $\approx$
A) 0.05
B) 0.06
C) 0.075
D) 0.09
5. Camber correction is required when:
A) Straight road
B) Superelevation provided
C) Level terrain
D) High traffic
6. Road margin includes all EXCEPT:
A) Shoulder
B) Parking lane
C) Median
D) Carriageway
7. For a two-way cycle track (separate from carriageway), recommended width is about:
A) 1.5 m
B) 2.0 m
C) 2.5 m
D) 3.0 m
8. The main purpose of providing kerbs is to:
A) Increase pavement thickness
B) Separate carriageway from shoulder/footpath
C) Reduce superelevation
D) Control camber
9. Camber for gravel road $\approx$
A) 1 in 50
B) 1 in 40
C) 1 in 30
D) 1 in 20
10. Stopping sight distance depends on:
A) Speed only
B) Friction only
C) Speed & friction
D) Width only

Quiz 1- Set F, B.Tech 4th Sem, GCU
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Roll No. _____

1. The second 20-year road plan was:
A) Nagpur
B) Bombay
C) Lucknow
D) Jayakar
2. VR (Village Roads) fall under:
A) Primary system
B) Secondary system
C) Tertiary system
D) Expressways
3. Trial pits are dug during:
A) Reconnaissance
B) Preliminary
C) Final location
D) Traffic survey
4. For $V=70$ km/h, $R=200$ m, $e \approx$
A) 0.05
B) 0.06
C) 0.07
D) 0.08
5. For a single-lane road with carriageway width 3.75 m and shoulders 1.5 m each side, roadway width equals:
A) 3.75 m
B) 5.25 m
C) 6.75 m
D) 7.50 m
6. For crest vertical curve, sight distance depends on:
A) Friction
B) Height of eye & object
C) Camber
D) Width
7. The main purpose of shoulder is:
A) Drainage only
B) Emergency stopping
C) Decoration
D) Speed increase
8. For a two-way cycle track (separate from carriageway), recommended width is about:
A) 1.5 m
B) 2.0 m
C) 2.5 m
D) 3.0 m
9. Maximum superelevation generally limited to:
A) 4%
B) 7%
C) 10%
D) 12%
10. The minimum width of a parking lane parallel to kerb for cars is generally:
A) 1.8 m
B) 2.2 m
C) 2.5 m
D) 3.0 m